

Heterogeneous Integrated III-V-on-SOI Transmitter for 6G FiWi mmWave/FSO Integrated Sensing and Communication

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Abstract: A heterogeneous fully integrated photonics transmitter with a III-V-on-SOI laser and EAM is experimentally presented in a Fi-Wi and FSO links, achieving 16-QAM transmission at 24 Gbps capacity and a LFM CW with a resolution of 0.03 m at 384 m radar range for emerging 6G ISAC networks.

Introduction

The deployment of 5G for Radio Access Networks (RANs), the use of different Frequency Range (FR), including FR1 (0.4 – 7 GHz), FR2 (24 – 70 GHz) and FR3 (7 – 24 GHz), and the demand for continuous, ultra-high data rate beyond the current 24.33 Gbps of CPRI/e-CPRI Rate-10 and low latency, down to <1 ms, in various applications have necessitate efficient utilization of the limited spectrum, and massive deployment of energy-efficient devices [1, 2]. In parallel, RANs are evolving towards a new era of concurrent environmental ‘sensing’, introducing Integrated Sensing and Communication (ISAC) functions in 6G, which are standardized by NextG, 6G-IA, ETSI and 3GPP alliances [3] – [6]. Recent roadmaps suggest steadily more stringent localization-sensing KPIs and Use cases, such as accuracy of 0.2 m – 0.5 m for human movement recognitions to airborne-environmental sensing and automotive scenarios, 100 ms sweeping-cycle times for traffic coordination or collision avoidance or 0.375 m resolution for gesture recognition and motion monitoring [2]. Such ISAC applications request precise localization of a User Equipment (UE) or distance object, that has been so far performed in lidars/radars, stimulating a strong interest in sharing hardware with conventional communications to facilitate joint ISAC transmitters with reduced cost, weight and size [5][6], especially as 6G scale in the vertical dimension with battery operated UAVs, drones or low/high altitude platform stations as depicted in

Fig. 1(a).

Global research efforts have focused on developing transmitters capable of high sensing resolutions with support for communications at all 6G FRs. The demonstrations of ISAC with electronic transmitters are usually faced with the well-known bandwidth and high-power consumption limitations [13], electro-optics transmitters used for ISAC demonstration in Radar [1][2][7][14] and LiDAR [8] applications have relied on external laser sources and other discrete devices [6] leading to bulky assemblies. This is due to the long-standing challenge of integrating laser and other III-V active components on Si platform [9], marking the distinct absence of a fully integrated PIC transmitter that meets all ISAC KPIs.

This work draws from latest heterogeneous integrations of compact III-V-on-SOI PICs for telecom [10], and spectroscopy [11], presents a fully-integrated PIC that utilizes a single, “unified” III-V epitaxy both for the laser and the modulator, achieving a compact, cost-efficient and fully-integrated solution with a good trade-off between optical power, electro-optics bandwidth and driving voltage. The PIC is experimentally tested in mmWave Fiber Wireless (Fi-Wi) and Free Space Optical (FSO) setups for three distinct 6G ISAC scenarios, achieving up to 24 Gbps data modulation using a 16-QAM format, operation at all FRs, resolution down to 0.03 m at a dynamic range up to 384 m and 2.54 μ s sweeping cycle in sensing scenarios. Finally, the PIC is benchmarked in a joint ISAC demonstration,

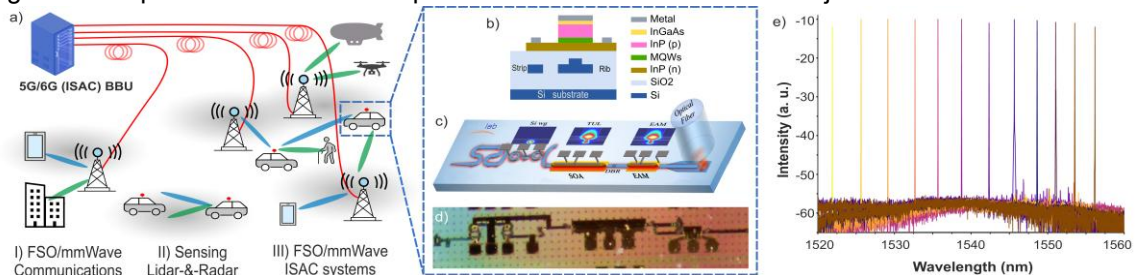


Fig. 1: a) Conceptual Schematic of the deployment of ISAC PIC for Fiber Wireless mmWave and free space optics (FSO) in 6G RANs communication. BBU –backbone unit. b)-d) Cross-section, schematic and photo of the developed ISAC PIC transmitter and e) spectrum of the tunable laser

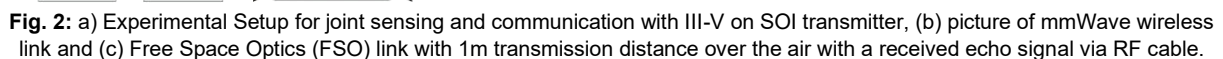
loss of 4 dB and a static extinction ratio of 7 dB under 2.5 V reversed bias voltage was measured, confirming its low-power consumption features in uncooled conditions.

The optical output from the PIC was obtained via grating coupler and propagated to the Fi-Wi setup, as shown in Fig. 2(a). Specifically, a Keysight N8195A AWG is used to generate various ISAC waveforms, with 2048 symbols of 16-QAM at FR1, 2 and 3 bands and LFM CW of Radar/LiDAR application, in order to modulate the EAM and imprint them on the optical carrier. The Analog Radio over Fiber (ARoF) was propagated through a 7 km single mode fiber spool, amplified by an erbium doped fiber amplifier to compensate the losses and spatially routed by an array waveguide grating-demultiplexer with of 0.7 nm channel bandwidth towards the FSO/mmWave stages.

The Over the Air (OTA) distances of the FSO and mmWave wireless links were 1 m and connected to different wavelengths output (λ_2 and λ_1) of the demultiplexer respectively. For the FSO link, two Thorlabs C80APC-C achromatic fiber collimated lens with 80 mm focal length were mounted on kinematic rotation stages and aligned across a 1 m link distance, while for the mmWave link, the signal was optical-electrical converted by a Lab Buddy InGaAs photoreceiver with 0.82 A/W responsivity and then a pair of Gotmic V-band Tx-Rx antennas (gMRX0061A) with 6x RF upconversion stages using a 16 dBm local oscillator signal of 10 GHz. Finally, the received 16-QAM and LFM CW signals were connected to a Real Time Oscilloscope (RTO) Keysight DSOZ632A for evaluation purpose.

Experimental Results

The PIC transmitter is benchmark in three transmission scenarios and use cases; B5G/6G communications, LiDAR/Radar sensing and finally, a joint ISAC scenario. In the first B5G/6G Fi-Wi ARoF mmWave/FSO fronthaul scenarios, two signals were synthesized and tested with 2048 symbols. Particularly, a 16-QAM signal at 2.5 GHz IF and a symbol rate of 1 Gbaud was transmitted OTA at 62.5 GHz by the horn antennas, while a 16-QAM signal at a 10 GHz IF and 6 Gbaud symbol rate was transmitted across



the FSO link. The received constellation diagrams are depicted in Fig. 3(b) and (d), featuring Error Vector Magnitude (EVM) of 12.035 % and 12.105 % respectively at a received optical power of -3 dBm and -5 dBm at the photodiode. This satisfies the 3GPP threshold values, thus confirming the capability of the PIC to transport user data of 4 Gbps and peak traffic capacities up to 24 Gbps respectively. EVMs measured over a range of received optical powers plotted in Fig. 3(c) show higher path loss in the Fi-Wi link.

Subsequently, towards radar sensing, a sawtooth LFM CW signal was synthesized centered at 5 GHz IF. The modulating LFM CW signal was swept across a 5 GHz bandwidth at a sweeping-cycle period of $2.56 \mu\text{s}$ with its generated Fourier response depicted in Fig. 3(e) revealing a flat frequency response and thus linear amplitude modulation across 2.5 GHz and 7.5 GHz. The signal was transmitted across the mmWave horn antennas, and to emulating the echo-signal reflection, Tx antenna was focus towards a static metallic object, placed 20 cm ahead of the antenna, which reflect the signal towards the Rx, the received signal is thus combined with a copy from the transmitted signal before feeding into the RTO. The time trace is plotted in Fig. 3(f) featuring a clearly swept and modulated signal ranging from -15 mV to $+15$ mV. Moreover, the waterfall spectrogram of the Tx and Rx LFM CW radar signals are shown in Fig. 3 (g), confirming a clear evolution of the frequency-versus-time sweeping-evolution with frequency modulation power between -135 dBm/Hz and -35 dBm/Hz. The beating frequency of the Tx-Rx radar-signal copies exhibits a range resolution down to only 0.03 m at an object-distance range of 384 m, that meet the respective 3GPP KPIs for automotive scenarios, detection and anthropomorphic motion-detection.

Finally, an ISAC scenario is evaluated where a sensing and a communication FR3 mmWave

signal were multiplexed in the frequency domain and simultaneously modulated by the PIC and transmitted over the FSO link. The linear triangle FMCW sensing signal featured a center frequency f_1 of 25 GHz swept at 1.25 GHz bandwidth at a sweeping-cycle period of $1.28 \mu\text{s}$, while the 16-QAM communication signal was centered at f_2 of 28 GHz with a symbol rate of 2 GBaud. The spectrum of the multiplexed signals is shown in Fig. 3(h), confirming a modulation bandwidth at least up to 30 GHz for mmWave frequencies and two-band signals. Furthermore, the constellation diagram of the received communication signal f_2 is clearly demodulated with an EVM of 11.5 % as shown in Fig. 3(i), corresponding to a 8 Gbps user data rate, while at the same time the spectrogram of the f_1 sensing signal shown in Fig. 3(j), corresponding to a sensing resolution down to only 0.75 m at a distance-range of 150 m, successfully confirming the capabilities of the PIC to simultaneously meet the respective 3GPP KPIs for 6G ISAC networks.

Conclusion

A compact, heterogeneous fully-integrated PIC transmitter with a III-V-on-SOI laser and EAM on a single Si-waveguide is experimentally presented, with at least 24 Gbps data rate using a 16-QAM, as well as a sensing resolution as low as 0.03 m at a range up to 384 m to meet the respective 3GPP KPIs of automotive vehicle object detection and anthropomorphic motion identification, shaping a promising roadmap for a low-cost, energy-efficient PIC solution for emerging 6G FiWi mmWave/FSO Integrated Sensing and Communications. While this experiment relies on external oscillator for upconversion, two tunable laser source with narrow linewidths could be realized on a similar integrated technology to enable the direct generation of mmWave signal on chip [12] for the ISAC transmission making the architecture more compact.

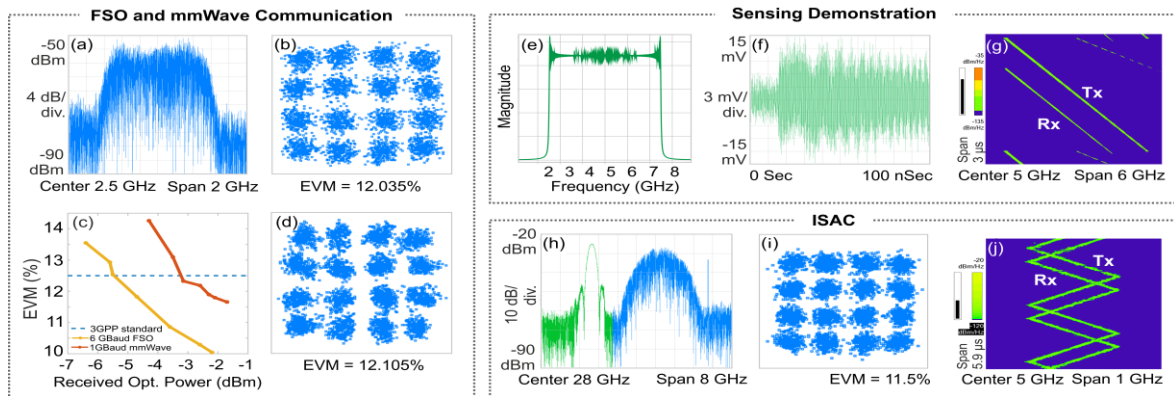


Fig. 3: Experimental results of PIC transmissions, including: (a)-(d) Communications over FiWi mmWave and FSO, (e)-(g) FFT, time-trace and spectrogram of sensing signal, (h)-(j) spectrum, constellation, spectrogram of FDM ISAC signals.

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